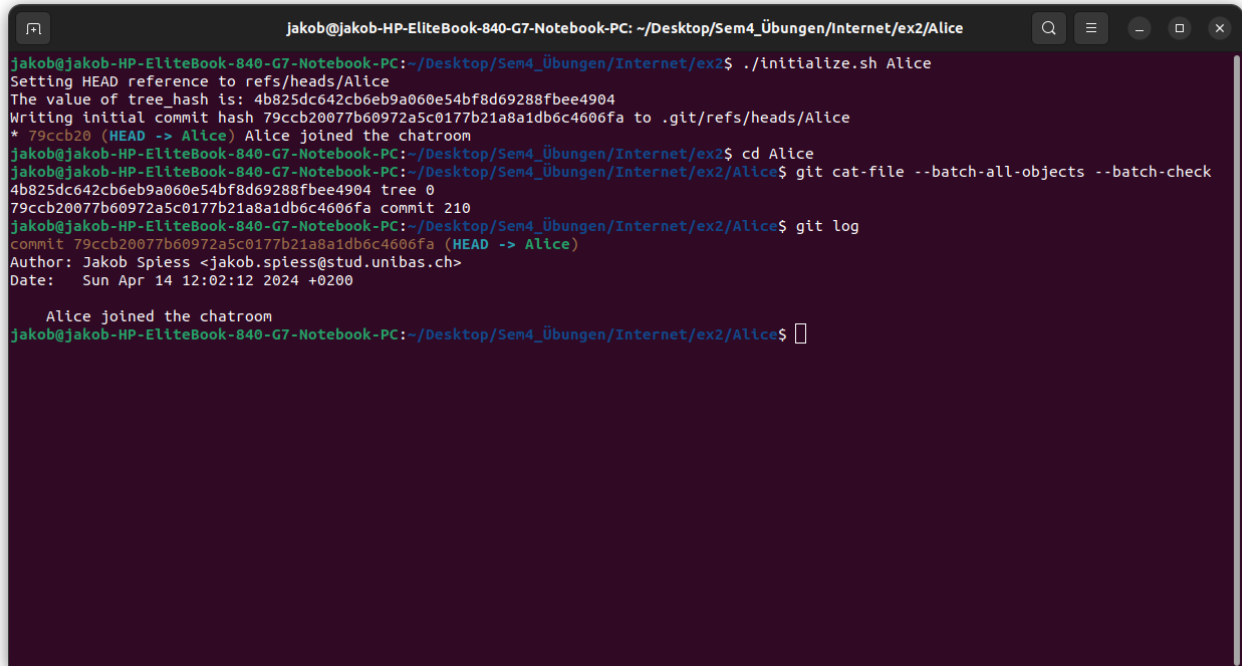


The following screenshots always show the terminal commands run and the terminal outputs for the exercises in Tasks 1-3. I always just run my scripts (initialize.sh, post.sh, show.sh, automate.sh) and the requested commands in the tasks. The source code files are also in the folder to look at the commented source code.

## Task 1 – Initialization

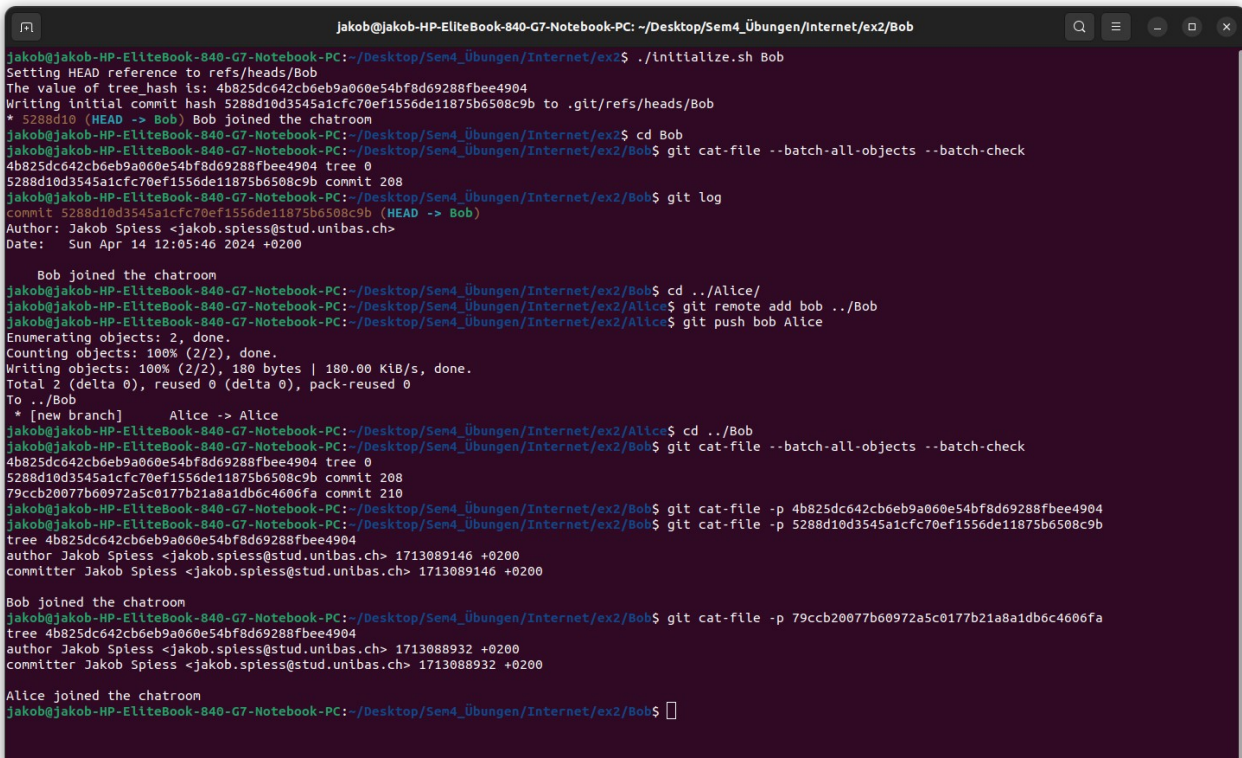
1b)



```
Jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4_Übungen/Internet/ex2/Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$ ./initialize.sh Alice
Setting HEAD reference to refs/heads/Alice
The value of tree_hash is: 4b825dc642cb6eb9a060e54bf8d69288fbee4904
Writing initial commit hash 79ccb20077b60972a5c0177b21a8a1db6c4606fa to .git/refs/heads/Alice
* 79ccb20 (HEAD -> Alice) Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$ cd Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git cat-file --batch-all-objects --batch-check
4b825dc642cb6eb9a060e54bf8d69288fbee4904 tree 0
79ccb20077b60972a5c0177b21a8a1db6c4606fa commit 210
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git log
commit 79ccb20077b60972a5c0177b21a8a1db6c4606fa (HEAD -> Alice)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:02:12 2024 +0200

    Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$
```

1c)



```
Jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4_Übungen/Internet/ex2/Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$ ./initialize.sh Bob
Setting HEAD reference to refs/heads/Bob
The value of tree_hash is: 4b825dc642cb6eb9a060e54bf8d69288fbee4904
Writing initial commit hash 5288d10d3545a1cfc70ef1556de11875b6508c9b to .git/refs/heads/Bob
* 5288d10 (HEAD -> Bob) Bob joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$ cd Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git cat-file --batch-all-objects --batch-check
4b825dc642cb6eb9a060e54bf8d69288fbee4904 tree 0
5288d10d3545a1cfc70ef1556de11875b6508c9b commit 208
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git log
commit 5288d10d3545a1cfc70ef1556de11875b6508c9b (HEAD -> Bob)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:05:46 2024 +0200

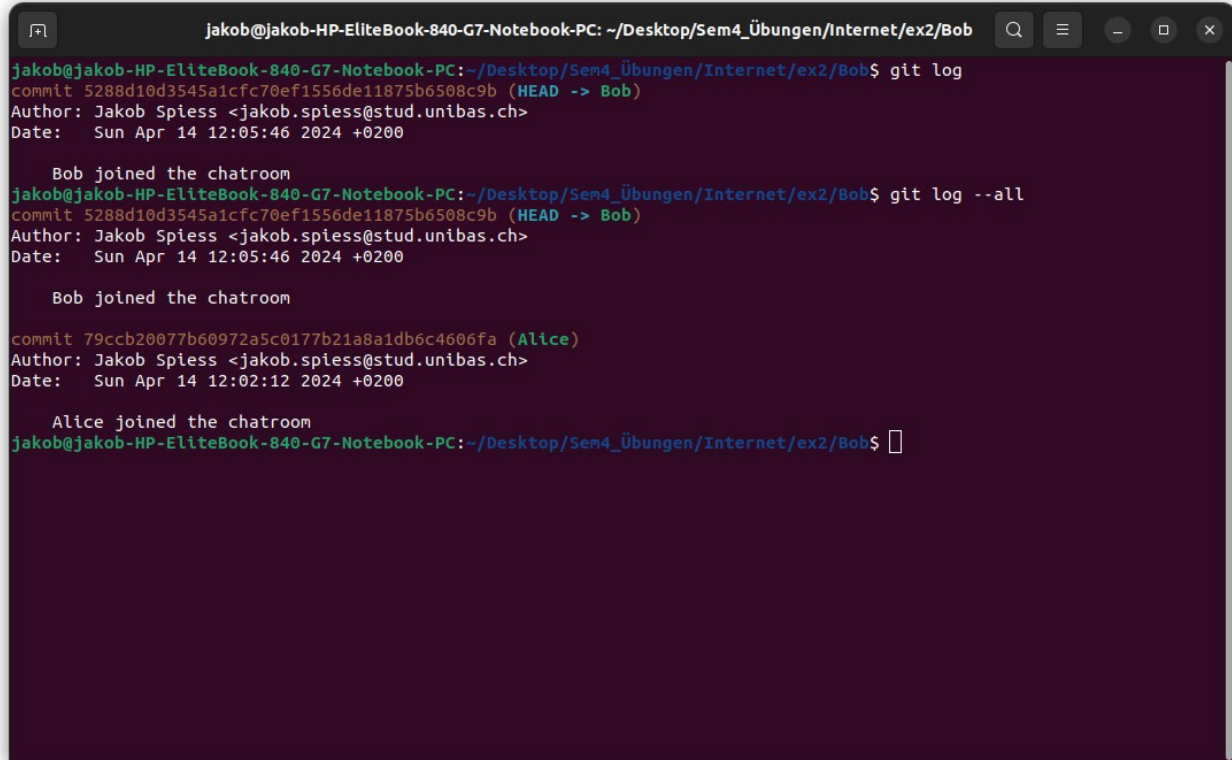
    Bob joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ cd ../Alice/
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git remote add bob ../Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git push bob Alice
Enumerating objects: 2, done.
Counting objects: 100% (2/2), done.
Writing objects: 100% (2/2), 180 bytes | 180.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0
To ../Bob
* [new branch] Alice -> Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ cd ../Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git cat-file --batch-all-objects --batch-check
4b825dc642cb6eb9a060e54bf8d69288fbee4904 tree 0
5288d10d3545a1cfc70ef1556de11875b6508c9b commit 208
79ccb20077b60972a5c0177b21a8a1db6c4606fa commit 210
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git cat-file -p 4b825dc642cb6eb9a060e54bf8d69288fbee4904
tree 4b825dc642cb6eb9a060e54bf8d69288fbee4904
author Jakob Spiess <jakob.spiess@stud.unibas.ch> 1713089146 +0200
committer Jakob Spiess <jakob.spiess@stud.unibas.ch> 1713089146 +0200

Bob joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git cat-file -p 79ccb20077b60972a5c0177b21a8a1db6c4606fa
tree 4b825dc642cb6eb9a060e54bf8d69288fbee4904
author Jakob Spiess <jakob.spiess@stud.unibas.ch> 1713088932 +0200
committer Jakob Spiess <jakob.spiess@stud.unibas.ch> 1713088932 +0200

Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$
```

1d)

The problem is that git log only shows the commit history for the branch currently checked out (<https://www.warp.dev/terminus/git-commit-history>). But we now have 2 branches in this repository with HEAD pointing to Bob, so it only shows the 1 commit on Bob's branch. At first I made a new commit and added Alice's one to Bob's one as a parent and saved this new one into .git/refs/heads/Bob, but since ordering doesn't have to be absolute I think this isn't necessary. Now I just ran git log -all to show commits across all branches:

A screenshot of a terminal window with a dark purple background. The window title is 'jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4\_Übungen/Internet/ex2/Bob'. The terminal shows the output of 'git log' and 'git log --all'. The first command shows a single commit for Bob. The second command shows two commits: one for Bob and one for Alice, with Bob's commit as the parent of Alice's. The commit messages are 'Bob joined the chatroom' and 'Alice joined the chatroom'.

```
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git log
commit 5288d10d3545a1cfc70ef1556de11875b6508c9b (HEAD -> Bob)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:05:46 2024 +0200

    Bob joined the chatroom

jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git log --all
commit 5288d10d3545a1cfc70ef1556de11875b6508c9b (HEAD -> Bob)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:05:46 2024 +0200

    Bob joined the chatroom

commit 79ccb20077b60972a5c0177b21a8a1db6c4606fa (Alice)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:02:12 2024 +0200

    Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$
```

## Task 2 – Posting Messages

```
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4_Übungen/Internet/ex2/Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ ../post.sh "Hello, everyone! I'm here."
parent_hashes_string: -p 79ccb20077b60972a5c0177b21a8a1db6c4606fa
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git push bob --all
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 222 bytes | 222.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To ../Bob
 79ccb20..84fd297 Alice -> Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ cd ../Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git log --all
commit 84fd297304dd37b9b5f33cbb3fc0eb4d4dcc5b2e (Alice)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:23:49 2024 +0200

    Alice (12:23:49): Hello, everyone! I'm here.

commit 5288d10d3545a1cfc70ef1556de11875b6508c9b (HEAD -> Bob)
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:05:46 2024 +0200

    Bob joined the chatroom

commit 79ccb20077b60972a5c0177b21a8a1db6c4606fa
Author: Jakob Spiess <jakob.spiess@stud.unibas.ch>
Date: Sun Apr 14 12:02:12 2024 +0200

    Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$
```

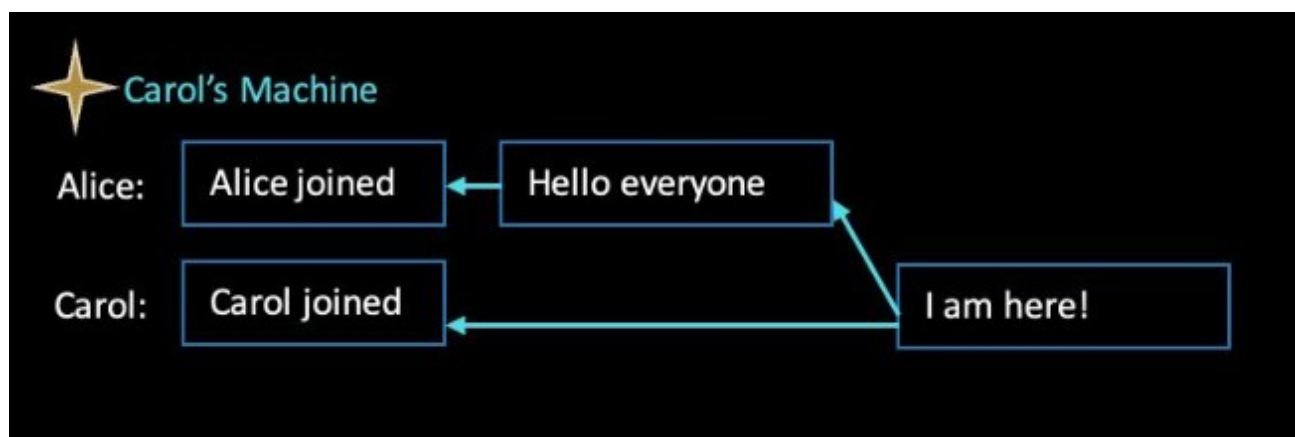
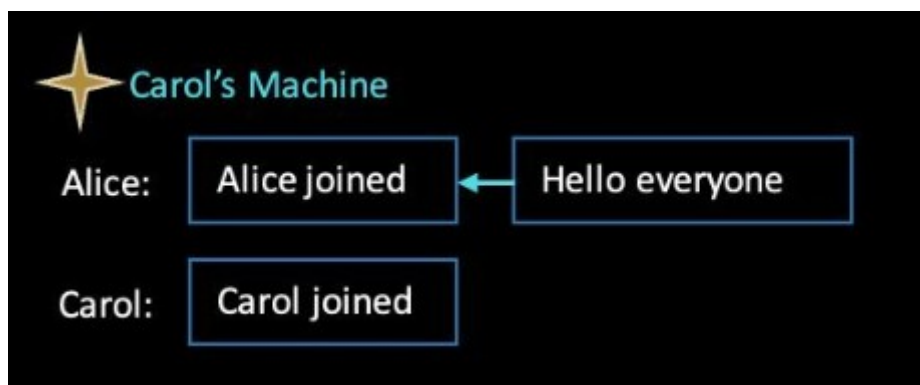
```
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4_Übungen/Internet/ex2/Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ ../post.sh "Hello, Alice! I'm new here too"
parent_hashes_string: -p 84fd297304dd37b9b5f33cbb3fc0eb4d4dcc5b2e -p 5288d10d3545a1cfc70ef1556de11875b6508c9b
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git add remote alice ../Alice
fatal: ../Alice: '../Alice' is outside repository at '/home/jakob/Desktop/Sem4_Übungen/Internet/ex2/Bob'
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git remote add alice ../Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ git push alice --all
Enumerating objects: 2, done.
Counting objects: 100% (2/2), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 317 bytes | 317.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
To ../Alice
 * [new branch]      Bob -> Bob
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Bob$ cd ../Alice
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git log --oneline --graph
* 84fd297 (HEAD -> Alice, bob/Alice) Alice (12:23:49): Hello, everyone! I'm here.
* 79ccb20 Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$ git log --oneline --graph Bob
* 40802f6 (Bob) Bob (12:28:03): Hello, Alice! I'm new here too
| \
| * 5288d10 Bob joined the chatroom
| * 84fd297 (HEAD -> Alice, bob/Alice) Alice (12:23:49): Hello, everyone! I'm here.
* 79ccb20 Alice joined the chatroom
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2/Alice$
```

### Task 3 – User Interface

3b)

```
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC: ~/Desktop/Sem4_Übungen/Internet/ex2
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$ ./automate.sh
Alice joined the chatroom
Alice (12:57:06): Hello, everyone! I'm here.
Bob joined the chatroom
Bob (12:57:06): Hello, Alice! I'm new here too.
Alice (12:57:06): How are you Bob?
Bob (12:57:06): Good thx and you?
Alice (12:57:06): Doing fine.
jakob@jakob-HP-EliteBook-840-G7-Notebook-PC:~/Desktop/Sem4_Übungen/Internet/ex2$
```

3c)



In the slides 56 & 57 it is shown how it would look if Carol pulls Alice's repo and then posts a message (just here the message "I am here!" instead of "Hi all"). If Carol posted the message

without pulling Alice's repo, the commit with the message would only have 1 parent which is the commit that is Carol's initial join message.